2406ET129

Roll No. | | | | | | | | | | | | |

Course: B. Tech. Computer Engineering/
B. Tech. Computer Science and Engineering (AI/ML)
Subject: Digital Electronics, Subject Code: ETCS-104
Semester: II

Time: 03 Hours **Max Marks: 70**

Instructions to the Students:

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 02 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

Q.No.	SECTION –A (SHORT ANSWER TYPE QUESTIONS)	Marks
1. a.	Determine $(377)_{10}$ in Octal and Hexa-Decimal equivalent.	(2)
b.	What is grey code and mention its advantages. Give the Gray code for the binary number $(111)_2$.	(2)
c.	Convert the following Excess 3 numbers into decimal numbers. i. 1011 ii. 1001 0011 0111	(2)
d.	What is a karnaugh map? Interpret the limitations of karnaugh map.	(2)
e.	Construct OR gate using only NAND gates.	(2)
f.	Why is MUX called as data selector?	(2)
g.	Convert T Flip Flop to D Flip Flop.	(2)
h.	What is race around condition in flip-flops?	(2)
i.	Differentiate asynchronous and synchronous type counters.	(2)
j.	Which is faster TTL or ECL? Which requires more power to operate?	(2)

SECTION –B (LONG ANSWER TYPE QUESTIONS)

2. a) Convert the following numbers (10)

   i. $(163.789)_{10}$ to Octal number

ii. $(11001101.0101)_2$ to base-8 and base-4

iii. $(4567)_{10}$ to base2

iv. $(4D.56)_{16}$ to Binary

b) Subtract $(111001)_2$ from $(101011)_2$ using 1's complement?

3. Simplify the Boolean expression (10)

i. $\overline{\overline{A+B}+\overline{C}}$

ii. $\overline{\overline{A}+B+\overline{CD}}$

4. Derive the expression for SOP form for the following: (10)

$Y=\sum m\,(1,3,4,5,6,7,8,13,14)$

Implement the function using only NOR gates.

5. i. Design a full adder using NOR gates (10)

ii. Design a full subtractor using half subtractors.

6. i. Draw and explain the operation of Master-Slave JK flipflop. (10)

ii. Design a 5-bit ring counter and mention its applications.

7. i. Explain the realisation of JK flipflop from T flipflop. (10)

ii. Write short notes on SIPO and draw the output waveforms.

8. Write a short note on the following: (10)

i. A to D Converters.

ii. Asynchronous Counters

9. Design a CMOS inverter and explain its operation. Comment on its characteristics such as Fan-in, Fan-out power dissipation, propagation delay and noise margin. Compare its advantages over other logic families. (10)

===END OF PAPER===